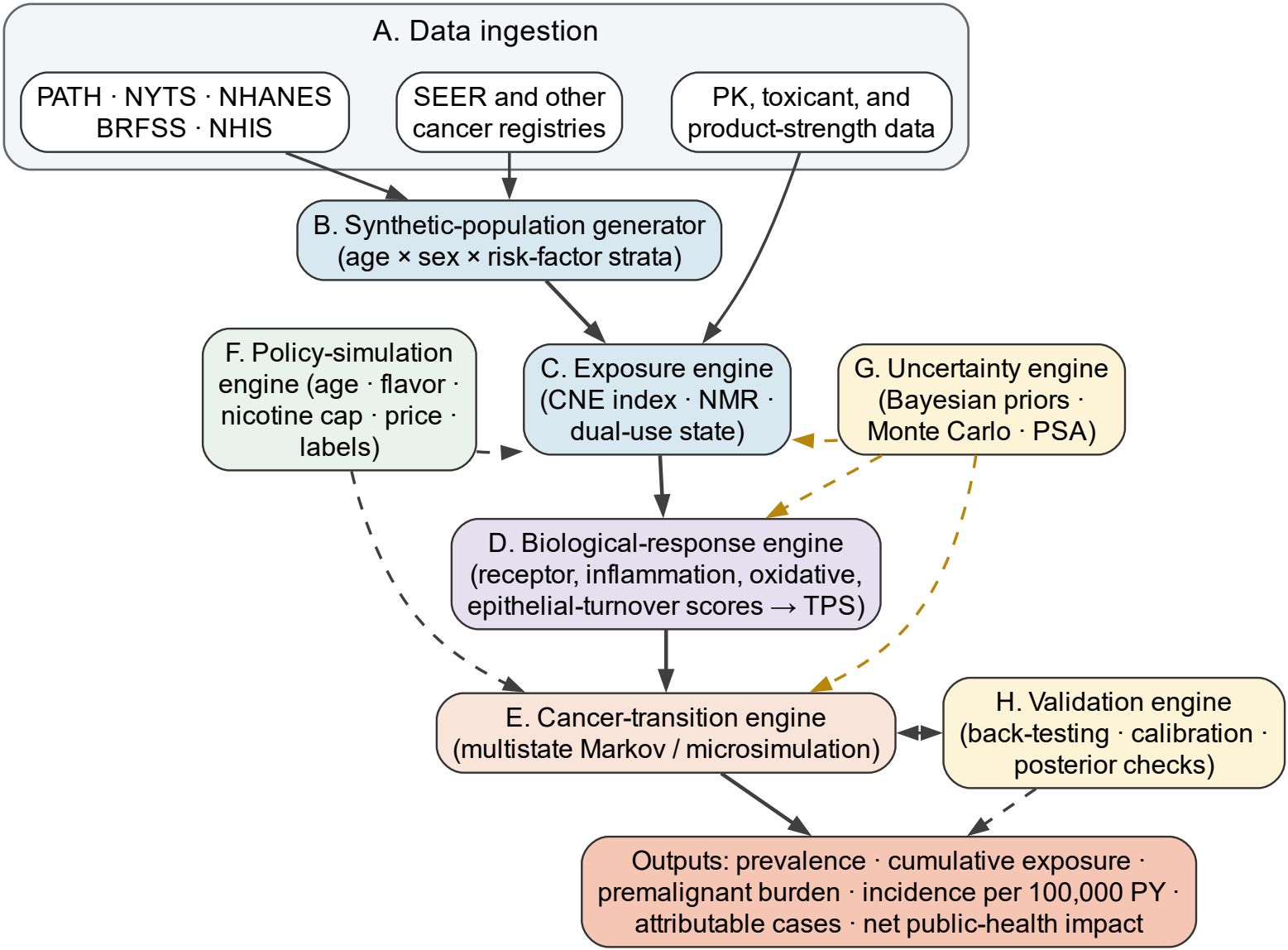
A. Data ingestion
PATH · NYTS · NHANES
BRFSS · NHIS
SEER and other
cancer registries
PK, toxicant, and
product-strength data
B. Synthetic-population generator
(age × sex × risk-factor strata)
F. Policy-simulation
engine (age · flavor ·
nicotine cap · price ·
labels)
C. Exposure engine
(CNE index · NMR ·
dual-use state)
G. Uncertainty engine
(Bayesian priors ·
Monte Carlo · PSA)
D. Biological-response engine
(receptor, inflammation, oxidative,
epithelial-turnover scores → TPS)
E. Cancer-transition engine
(multistate Markov / microsimulation)
H. Validation engine
(back-testing · calibration ·
posterior checks)
Outputs: prevalence · cumulative exposure ·
premalignant burden · incidence per 100,000 PY ·
attributable cases · net public-health impact